```
→ Pseudocode for Genetic Algorithm for optimization.

→ Input : POP_SIZE
  CROSSOVER_RATE, MUTATION_RATE , MAX_GEN
  Output: Best_solution

→ Define fitness (individual)

→ Population ← random individuals (POP_SIZE)

→ FOR generation = 1 to MAX_GEN DO

  DO
→ Evaluate fitness of each individual

→ Update best_solution if a better one is found

→  new_population ← empty

→ while size(new_population) < POP_SIZE DO

→ Parent1, parent2 ← select (population)

→  if random () < CROSSOVER_RATE Then

→ Child1, Child 2 ← CROSSOVER (parent1, parent2)

→ Else

→  child1, child2 ← parent1, parent2

→  END IF
→ MUTATE (CHILD1), MUTATE (child2) with MUTATION
  RATE.

→ ADD child1, child2 to new_population

→ End while

→ Population ← new_population

→ End for

→ Return best_solution

→ END
```

Cyber Attack Detection Optimization

Select features and tune model parameters for Network Intrusion detection.

Obj) Improve Detection performance by reducing false alarm and computational cost.

```
Algorithm Cyber attack_Detection_Optimization
Input: pop_size, max_gen, crossover_rate, mutation_rate,
Data
Output: best_features, best_params

POP ← init_population (pop_size)
        for generation ← 0 to max_gen - 1 do
                for each individual in pop do.
```

```
feat ← Extract Feature (individual .bits)
model ← Train (Data.Train, Feat, Param)
Evaluate fitness

new_pop = [ ]
while size (new.pop) < pop size
Parent1, Parent2 ← Select (pop)
if (random () < crossover_rate) then
    C1, C2 ← Crossover (Parent1, Parent2)

else
    C1, C2 ← Parent1, Parent2

    Mutate C1, C2 with mutation_rate
    add C1, C2 do new_pop
    pop = new_pop
    gen = gen + 1
Best ← getBest (pop)
return Best.bits, Best.values
```

→

Using Genetic Algorithm inspired by natural Selection improves Cyber Security and Intrusion detection System by automating feature selection and tuning the model by optimizing using best parameter available.

Output:

```
Gen.  Best Fitness  Features Selected  n_estimators  max_depth
1     0.7707        16                 120           9
2     0.7753        17                 50            9
3     0.7849        16                 70            24
4     0.7658        9                  110           18
5     0.7702        11                 50            24
6     0.7631        12                 110           21
7     0.7662        20                 70            21
8     0.7722        17                 70            21
9     0.7805        19                 110           12
10    0.7830        18                 130           18

Cyber Attack Detection optimization Complete

Optimal Fitness score : 0.7707
Optimized Hyperparameters: n_estimators = 150, max_depth = 9
Selected Cyber features (16/20): ['src_bytes', 'dst_bytes',
'urgent', 'hot', 'num_failed_logins', 'num_comprimised', 'root_shell
'is_guest_login', 'count', 'rerror_rate', 'same_srv_rate',
'srv_diff_host_rate', 'dst_host_srv_count', 'dst_host_same
srv_rate', 'dst_host_same_src_port_rate'].
```